

Summer School of Dynamical Systems

Nanjing University

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This note selected materials from the Summer School held by Nanjing University from June 29 to July 12, 2025. The program consisted of a series of short courses and research lectures covering various topics in dynamical systems. Among these, two courses stood out for their depth and relevance to my interests; I provide detailed notes on both. Additionally, I include selected concepts and ideas gathered from other lectures, with the aid of AI tools for summarization and clarification. An appendix lists the associated files and supplementary materials.

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1 胡俊

Topic Dynamics of quadratic and cubic rational maps

Abstract I will give very brief introduction to the dynamical systems generated by quadratic or cubic rational functions. Included topics are: period-doubling bifurcation, renormalization, Feigenbaum rigidity, entropy, boundary of chaos, classification of Fatou sets or Julia sets, Mandelbrot sets, and some related open problems.

This is actually an overview of a 56 teaching hours course.

1.1 Sarkovskii's Theorem

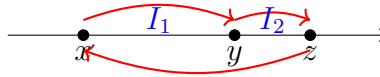
Exercise Any continuous map from a closed finite interval into itself has at least one fixed point.

Exercise Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and $f[a, b] \subset [a, b]$, then f has at least one fixed point in $[a, b]$.

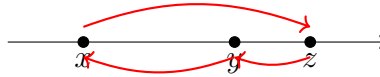
Thm (Special case of Sarkovskii's thm) Period 3 implies all periods. More specifically, let I be a nonempty connected subset of $\mathbb{R}^1$, i.e. $I = [a, b]$ or $[a, b)$ or $(a, b]$ or (a, b) , $a, b \in \mathbb{R}^1 \cup \{\infty\}$. Let $f : I \rightarrow \mathbb{R}$ be continuous. If f has a periodic point of period 3, then $\forall n \in \mathbb{N}$, f has a n -periodic point.

Proof Let (x, y, z) be a 3-cycle, we may assume that $x < y < z$. There are two cases

Case I



Case II



Without loss of generality, we consider case I. Then

$$I_1 \subset f(I_1), I_1 \cup I_2 \subset f(I_2).$$

$I_2 \subset f(I_2)$, then $\exists$ fixed point $p \in I_2, p \neq y, z$. Therefore p is a 1-period point.

$I_2 \subset f(I_1), I_1 \subset f(I_2)$, then $I_1 \subset f^2(I_1)$, then $\exists$ fixed point of f^2 , say $q \in I_1, q \neq x, y$. Therefore q is a 2-period point.

For the remaining $n \geq 4$ case, consider cover chain

$$I_2 \rightarrow I_2 \rightarrow \cdots \rightarrow I_2 \rightarrow I_1 \rightarrow I_2$$

such that $I_2 \subset f^n(I_2)$, then $\exists$ fixed point of f^n , say p . If period of p is $m < n$, then $m|n$, so at least $m \leq \frac{1}{2}n$. However, this means the first m steps in the cover chain forms a cycle, and $f^{n-1}(p) \in I_2 \neq I_1$.

Def Sarkovskii's order on the positive integers:

$$\begin{aligned} & 3 \triangleright 5 \triangleright 7 \triangleright \cdots \\ & \triangleright 3 \times 2 \triangleright 5 \times 2 \triangleright 7 \times 2 \triangleright \cdots \\ & \triangleright 3 \times 2^2 \triangleright 5 \times 2^2 \triangleright 7 \times 2^2 \triangleright \cdots \\ & \quad \quad \quad \cdots \\ & \cdots \triangleright 2^3 \triangleright 2^2 \triangleright 2 \triangleright 1 \end{aligned}$$

Thm (General case of Sarkovskii's thm) Let I be a nonempty connected subset of $\mathbb{R}^1$, let $f : I \rightarrow \mathbb{R}^1$ be continuous. If f has a periodic point of period n , then $\forall n \triangleright m$, f has a m -periodic point.

Proof Refer to *An Introduction to Chaotic Dynamical Systems* by Robert Devaney.

1.2 Topological Entropy

Def Let X be a compact metric space and $f : X \rightarrow X$ be continuous. $\forall \epsilon > 0, n \in \mathbb{N}$, a subset S of X is said to be (n, ϵ) -separated if $\forall x, y \in S$,

$$\max_{0 \leq k \leq n} d(f^k(x), f^k(y)) \geq \epsilon.$$

Rmk X is compact so S is finite.

Def Denote by $H(f, n, \epsilon)$ the maximal cardinality of all (n, ϵ) -separated sets. The topological entropy of f is defined to be

$$h(f) = \lim_{\epsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \ln H(f, n, \epsilon).$$

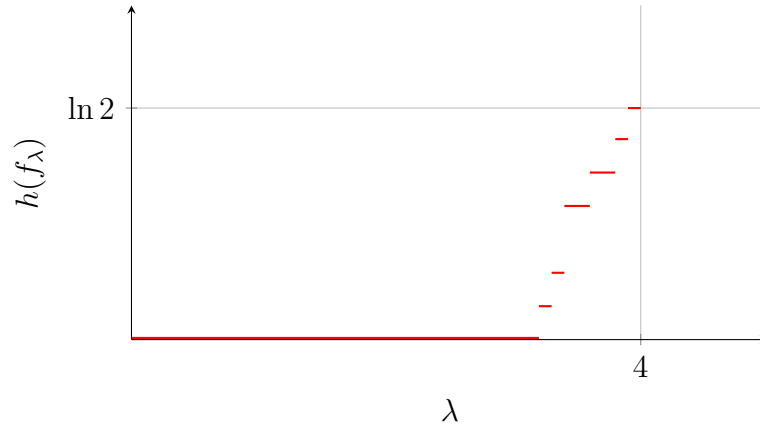
Fact I be a finite closed interval on $\mathbb{R}^1$, $f : I \rightarrow I$ be continuous. Then

$$h(f) = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \text{Var}(f^n).$$

Def Specifically, we consider $f_\lambda(x) = \lambda x(1 - x), f : [0, 1] \rightarrow [0, 1], \lambda \in [0, 4]$.

Prop $h(f_0) = 0$. $h(f_4) = \ln 2$.

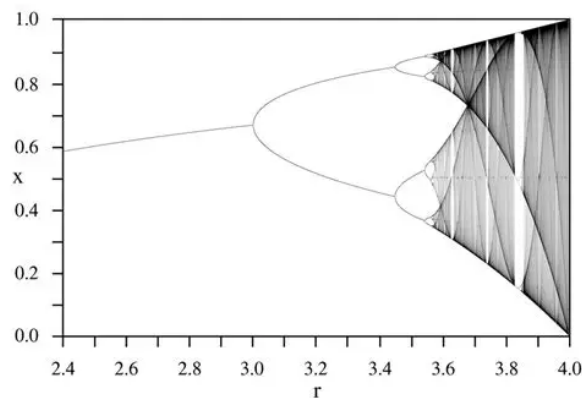
Thm Consider $h(f_\lambda)$ as a function of λ , then $h(f_\lambda) : [0, 1] \rightarrow [0, \ln 2]$ is monotone (by Milnor and Thurston) and continuous (relatively easy).



Rmk A very casual sketch is shown above. (Be careful!)

We call $\sup_{\lambda} \{h(f_\lambda) = 0\}$ the accumulation point of period-doubling bifurcation, it is between 3 and 4.

Note The entropy function is closely related to the bifurcation diagram of logistic map.



Thm (Bowen and Franks, 1976; Misiurewicz, 1979) Let $f \in C^0(I)$, then $h(f) > 0$ if and only if f has a periodic point whose period is NOT a power of 2.

Proof “ $\Leftarrow$ ” is relatively easy, we give a sketch. For example, f has a 3-periodic point, and the 3-cycle (x, y, z) acts as in case I. Then the transition matrix is

$$\begin{matrix} & I_1 & I_2 \\ \begin{matrix} I_1 \\ I_2 \end{matrix} & \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \end{matrix}$$

with eigenvalue $\frac{1 \pm \sqrt{5}}{2}$. The conclusion can be derived using the fact:

$$h(f) \geq \ln \text{maximal eigenvalue} = \ln \frac{1 + \sqrt{5}}{2} > 0.$$

“ $\Rightarrow$ ” is not easy, we omit it.

Cor If $h(f) = 0$, then all periodic points have period in the form of 2^n .

1.3 Period-Doubling Bifurcation, Feigenbaum Rigidity, Coulet-Tresser Rigidity

Def Let $f_\lambda = \lambda x(1 - x)$, $\lambda \neq 0$, then f_λ can be conjugated to $Q(x) = x^2 + r$ by an affine map $\varphi(x) = ax + b$ ($a \neq 0$)

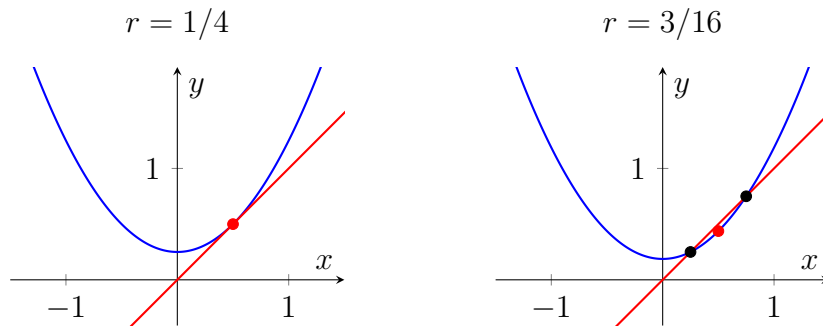
$$\begin{array}{ccc} \mathbb{R}^1 & \xrightarrow{f_\lambda} & \mathbb{R}^1 \\ \varphi \downarrow & & \downarrow \varphi \\ \mathbb{R}^1 & \xrightarrow{Q_r} & \mathbb{R}^1 \end{array}$$

And the numerical relation is

$$\begin{cases} a = -\lambda \\ b = \frac{1}{2}\lambda \\ r = \frac{\lambda^2}{4} - \frac{\lambda}{2} \end{cases}$$

Now we consider $Q_r(x)$.

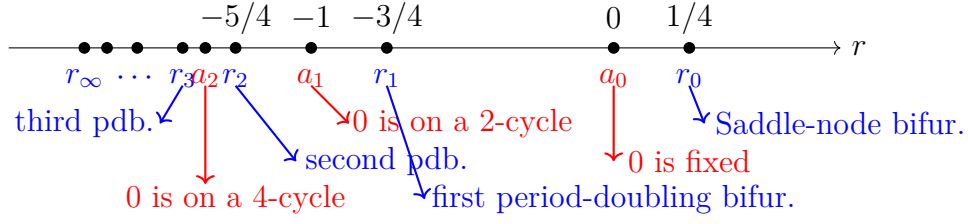
Def The unique fixed point bifurcates when r becomes smaller, and the previous fixed point turns attracting, this is called a saddle-node bifurcation. See figure below.



Fact Q_r has a saddle-node bifurcation at $r = \frac{1}{4}$.

Q_r has a period-doubling bifurcation at $r = -\frac{3}{4}$.

1.3.1 Properties on the r -axis



Def There is a sequence of period-doubling bifurcation parameters, denoted by $\{r_n\}_{n=0}^\infty$.

Prop

$$\lim_{n \rightarrow \infty} \frac{r_n - r_{n+1}}{r_{n+1} - r_{n+2}} = c \approx 4.6692 \dots$$

so the limit exists

$$r_\infty = \lim_{n \rightarrow \infty} r_n.$$

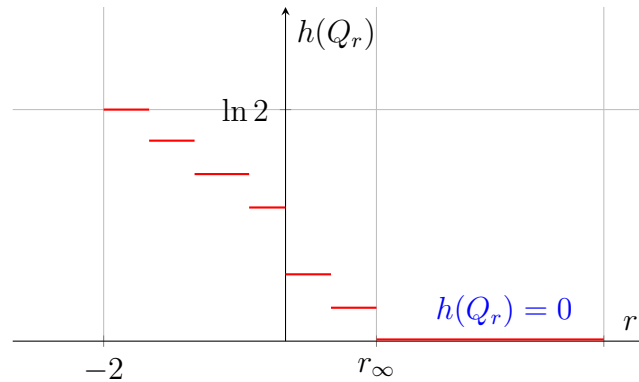
Fact There is a sequence of parameters that 0 is on a 2^n -cycle, denoted by $\{a_n\}_{n=0}^\infty$.

Prop Each a_n lies between r_n and r_{n+1} , so a_n is decreasing, with

$$\lim_{n \rightarrow \infty} a_n = r_\infty.$$

$$\lim_{n \rightarrow \infty} \frac{a_n - a_{n+1}}{a_{n+1} - a_{n+2}} = \lim_{n \rightarrow \infty} \frac{r_n - r_{n+1}}{r_{n+1} - r_{n+2}}.$$

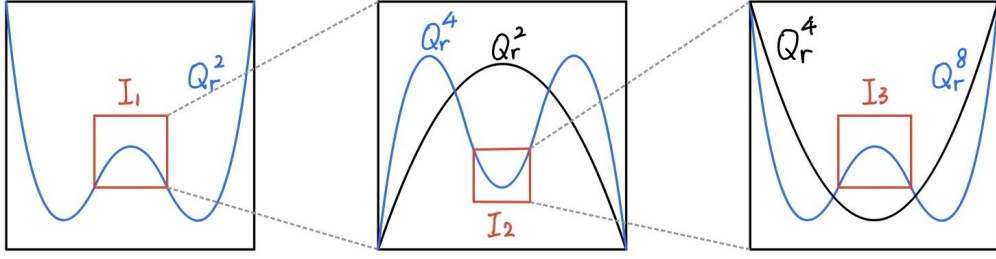
Fact r_∞ (related to Q_r) is conjugate to the accumulation point of period-doubling bifurcation (related to f_λ). So we can also draw the conjugated graph of topological entropy as



Note that the position of y -axis might be WRONG!

1.3.2 Renormalization

We roughly introduce it by graphs.



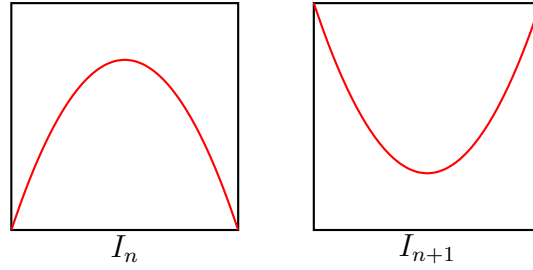
Each $Q_r^{2^n}|_{I_n}$ is locally quadratic-like

Def Let I_n be the domain of the n th renormalization of Q_{r_∞} , i.e.

$$Q_{r_\infty}^{2^n} : I_n \rightarrow I_n.$$

is a quadratic -like map.

Rmk $Q_{r_\infty}^{2^n}|_{I_n}$ may open upward or downward, but they are generally conjugated.



1.3.3 Properties of Q_{r_∞}

1 Let P denote the collection of periods, then

$$P(Q_{r_\infty}) = \{2^n, n = 0, 1, 2, \dots\}.$$

2 Every 2^n -cycle is simple, where simple means it is not composed of smaller repeated sub-cycles and consists of distinct points.

3 Q_{r_∞} is infinitely renormalizable.

4 The Couillet-Tresser rigidity is

$$\lim_{n \rightarrow \infty} \frac{I_{n+1}}{I_n} = c \approx 0.3995 \dots$$

5

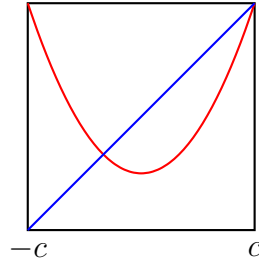
$$\bigcap_{n=1}^{\infty} \left(\bigcup_{k=0}^{2^n-1} Q_{r_\infty}^k(I_n) \right)$$

is a Cantor set, with Hausdorff dimension $\text{HD} \in (0, 1)$.

6 Denote the larger fixed point of Q_{r_∞} by c , then $\forall x \in [-c, c]$, the iteration of x under Q_{r_∞} converges to

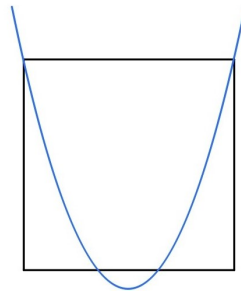
$$K = \overline{\{\text{The orbit of } 0 \text{ under } Q_{r_\infty}\}}.$$

K is called the Cantor set attractor of Q_{r_∞} .



1.3.4 Renormalizability

Def (informal) There is a sequence of b_n such that $Q_{b_n}^{2^n}|_{I_n}$ is just renormalizable.



a case when renormalization cannot work

1

$$2h(Q_{b_{n+1}}) = h(Q_{b_n}).$$

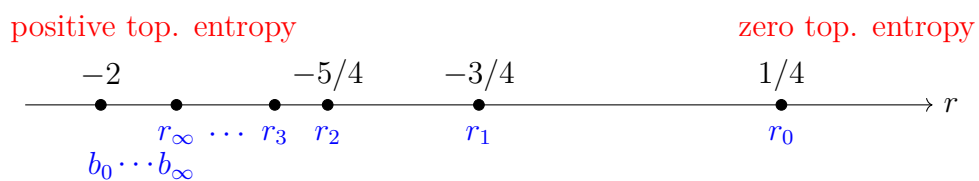
2 There exists

$$b_\infty = \lim_{n \rightarrow \infty} b_n.$$

3

$$P(Q_{b_\infty}) = \{2^n, n = 0, 1, \dots\}$$

4 Q_{b_∞} is infinitely renormalizable.



Question Is it true that $b_\infty = r_\infty$? Alternatively, is the following a single point set?

$$\left\{ r \in [-2, \frac{1}{4}], P(Q_r) = \{2^n, n = 0, 1, \dots\} \right\}$$

Thm (Sullivan, about 1993)

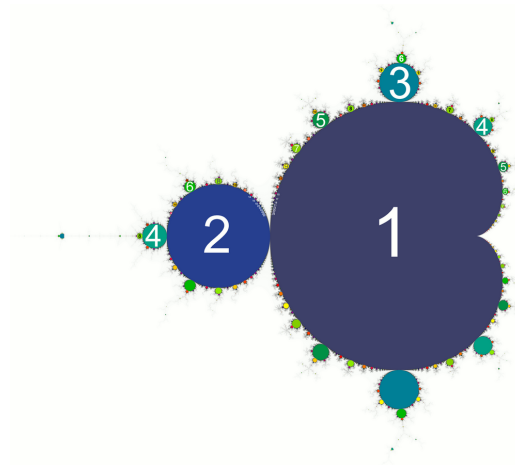
$$b_\infty = r_\infty.$$

Tools for proof Complex dynamics and Teichmüller theory.

1.4 Mandelbrot set

Def

$$M = \{c \in \mathbb{C}, Q_c^n(0) \text{ bounded as } n \rightarrow \infty\}.$$



Fact $c \in M \Leftrightarrow$ the filled Julia set K is connected.

$c \notin M \Leftrightarrow K$ is a Cantor set.

Def $c \in \mathbb{C}$ is a hyperbolic parameter if Q_c has an attracting cycle. Q_c is called a hyperbolic system.

Fact Let

$$\mathcal{H} = \{\text{hyperbolic parameters}\}$$

$$\mathcal{H}_0 = \{Q_c \text{ has an attracting fixed point}\}$$

$$\mathcal{H}_n = \{Q_c \text{ has an attracting } n\text{-cycle}\}$$

Then all $\mathcal{H}$ are open, $\mathcal{H}_n \subset \mathcal{H}$.

Def Q_c is called stable if $\exists \epsilon$, such that $\forall c'$ with $|c' - c| < \epsilon$, $Q_{c'}|_{J(Q_c)}$ is conjugate to $Q_c|_{J(Q_c)}$.

Def A parameter is called unstable if there exists a neutral periodic orbit.

Fact If Q_c is hyperbolic, then Q_c is stable.

Open Problem Is the converse true? precisely, if Q_c is stable, is Q_c hyperbolic?

This can be derived by the following.

Open Problem Is M locally connected?

Rmk Partial exploration: Bifurcation points are locally connected (by Lyubich).

Preliminary exploration: Is Julia set locally connected?

Thm (Yoccoz, about 1990) If Q_c is not infinitely renormalizable, then $J(Q_c)$ is locally connected.

Thm (胡俊, 蒋云平, 1998) $J(Q_{r_\infty})$ is locally connected.

Rmk The Julia sets of many infinitely renormalizable quadratic polynomials are proved to be locally connected.

1.5 Other topics

1.5.1 Herman ring

Thm 1 (Generally known) Polynomials have no Herman ring.

Thm 2 (Shishikura, 1987) Quadratic rational functions $Q(z) = \frac{az^2 + b}{cz^2 + d}$ ($ad \neq bc$) has no Herman ring.

Thm 2' (Milnor, 2004) Generalize Thm 2 to $Q(z) = \frac{az^n + b}{cz^n + d}$ ($n \geq 2$).

Thm 3 (胡俊, 肖映青, 2019) Totally ramified rational maps have no Herman ring, where totally ramified means every preimage of every critical value is a critical point.

Thm 3 can derive both Thm 2 and 2'.

1.5.2 Julia set Dichotomy

The Julia set of $Q(z) = \frac{az^n + b}{cz^n + d}$ is either connected or a Cantor set.

1.5.3 Cubic case

	Cubic Poly.	Quadratic Rational	Cubic Rational
Critical Points	2	2	4
Dim. of Para. Space	2 complex	2 complex	4 complex

2 崔巍巍

Topic 有限型函数逃逸集的 Hausdorff 维数

Abstract 有限型函数是超越动力系统的一类重要函数, 其在动力系统与函数论等多个领域均具有重要的作用. 在本次短课程中, 我们主要介绍近年来在有限型函数所生成的逃逸集的 Hausdorff 维数取值问题上的相关研究.

Refer to *Complex Dynamics II* by Bergweiler and the following two papers.

Hausdorff dimension of escaping sets of Nevanlinna functions

Hausdorff dimension of escaping sets of meromorphic functions

2.1 Escaping set

Consider $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ as transcendental entire function (TEF) or meromorphic function. We mainly focus on Eremenko-Lyubich class and Speiser class.

Def (Singular value) Denoted by $Sing(f)$, consists of

Asymptotic values, for example $0, \infty$ of e^z .

Critical values: if $f^\#(z) = 0$, then $f(z)$ is cv. Here $f^\#(z)$ is the spherical derivative, in order to deal with poles.

Def (class B and class S) $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ transcendental.

f is in Eremenko-Lyubich class if $Sing(f) \cap \mathbb{C}$ is bounded (i.e. omit ∞ as singular value), we call it class B .

f is in Speiser class if $Sing(f)$ is finite, we call it class S .

Prop

$$S \subset B.$$

Examples

- $f(z) = \sin(z)$. $Sing(f) = \{\pm 1, \infty\}$, so $f \in S$.
- $f(z) = ze^z$. $Sing(f) = \{0, -1/e, \infty\}$, so $f \in S$.

- $f(z) = z^2 e^z$. $Sing(f) = \{0, 4/e^2, \infty\}$, so $f \in S$.
- $f(z) = p(z)e^{q(z)}$, where p, q are polynomials, then $f \in S$.
- Elliptic function, for example the Weierstrass $\wp$ function.

$$Sing(\wp) = \{\wp(0), \wp(1/2), \wp(\pi/2), \wp(1/2 + \pi/2)\}$$

so $\wp \in S$.

- $f(z) = \sin(z)/z$, then $f \in B \setminus S$.

Rmk We will need Teichmüller space from function theory.

Class S has no wandering domain (Sullivan), but class B may have. Actually class S acts like polynomials or rational functions.

Def (Escaping set)

$$I(f) = \{z \in \mathbb{C}, f^n(z) \rightarrow \infty \text{ as } n \rightarrow \infty\}.$$

Thm (McMullen, 1987)

$$I(\lambda e^z) \neq \emptyset.$$

In fact we can find points of any escaping speed by constructing nesting sets.

Thm (Eremenko, 1989) f transcendental, then

$$I(f) \cap J(f) \neq \emptyset.$$

Thm f meromorphic, then

$$f \in B \Rightarrow I(f) \neq \emptyset.$$

2.2 Tool Theorems

2.2.1 Quasiconformal mapping

The first theorem below is to construct a q.c. meromorphic function. The second theorem tells when is a q.c. close to being conformal, so that we can control the deformation.

Def (Beltrami equation)

$$f_{\bar{z}} = \mu f_z.$$

where μ is called deformation function or Beltrami coefficient.

A map f is conformal if $f_{\bar{z}} = 0$, i.e. $\mu \equiv 0$; and is quasiconformal if $|\mu(z)| < 1$ almost everywhere.

Thm (Measurable Riemann Mapping theorem) Given μ satisfying $|\mu| \leq k < 1$, then there exists a q.c. $\phi : \mathbb{C} \rightarrow \mathbb{C}$ that $\mu_\phi = \mu$ a.e.

Specifically, for any $\mu : \mathbb{C} \rightarrow \mathbb{C}$ measurable and $\|\mu\|_\infty < 1$, there exists a quasiconformal $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfies the Beltrami equation, i.e.

$$f_{\bar{z}} = \mu f_z \quad (\text{in distributional sense})$$

with the normalization conditions: $f(0) = 0$, $f(1) = 1$, $f(\infty) = \infty$.

Moreover, this solution is unique under these normalizations.

Thm (Teichmüller-Wittich-Belinskii theorem) ϕ is q.c., let

$$K_\phi(z) = \frac{1 + |\mu_\phi|}{1 - |\mu_\phi|}$$

If

$$\iint_{|z| \geq 1} \frac{K_\phi(z) - 1}{|x|^2 + |y|^2} dx dy < \infty$$

Then $\phi(z) \sim z$ as $z \rightarrow \infty$.

Rmk $K_\phi(z)$ also measures deformation. Since we investigate $I(f)$, we only care about the neighborhood of ∞ , and this theorem tells how ϕ acts as $z \rightarrow \infty$.

2.2.2 Nevanlinna theory

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an ETF. Let

$$M(r, f) = \max_{|z|=r} |f(z)|.$$

Def (Growth order)

$$\rho(f) = \limsup_{r \rightarrow \infty} \frac{\log \log M(r, f)}{\log r}.$$

Example $\rho(e^z) = 1$.

Rmk Meromorphic functions don't have growth order, since they have poles.

Def For transcendental meromorphic functions, define

$$n(r, f, a) = \#\{f(z) = a, z \in D(0, r)\}.$$

Usually we denote $n(r, f) = n(r, f, \infty)$.

Lemma (Teichmüller) $f \in B$, if ∞ is not asymptotic value, then

$$\rho(f) = \limsup_{r \rightarrow \infty} \frac{\log n(r, f)}{\log r}.$$

i.e. growth order is decided by number of poles.

2.3 Preview results

2.3.1 Entire case

- McMullen 1987: $\dim_H I(\lambda e^z) = 2$.
- Barański 2008, Schubert 2007(?): $f \in B, \rho(f) < \infty \Rightarrow \dim_H I(f) = 2$.
- Bergweiler-Karpinska-Stallard 2009: $f \in B$ with *small* infinite order $\Rightarrow \dim_H I(f) = 2$.
- Rempe-Stallard 2009: $f \in B \Rightarrow \dim_H I(f) = [1, 2]$.

They use Cauchy integral when constructing $\dim_H I(f) = 1$.

2.3.2 Mero case

- 邱维元 1994: f be a Nevanlinna function and ∞ is asymptotic value $\Rightarrow \dim_H I(f) = 2$.
- Bergweiler-Rippon-Stallard 2009: $f \in B, \rho(f) < \infty, \infty$ is asymptotic $\Rightarrow \dim_H I(f) = 2$.
- Bergweiler-Kotus 2012: $f \in B, \rho(f) < \infty, \infty$ is NOT asymptotic $\Rightarrow$

$$\dim_H I(f) \leq \frac{2M\rho(f)}{2 + M\rho(f)}.$$

Here M is the supremum of the order of poles. The upper bound is usually rigorous, but not for some special examples.

- 崔巍巍 2021: f Nevanlinna function, ∞ is NOT asymptotic $\Rightarrow$

$$\dim_H I(f) = \frac{p+2}{p+4}, p = \deg S_{f(z)}.$$

- Bergweiler-Kotus 2012: $f \in B \Rightarrow \dim_H I(f) = [0, 2]$.

Compare: entire case is $[1, 2]$.

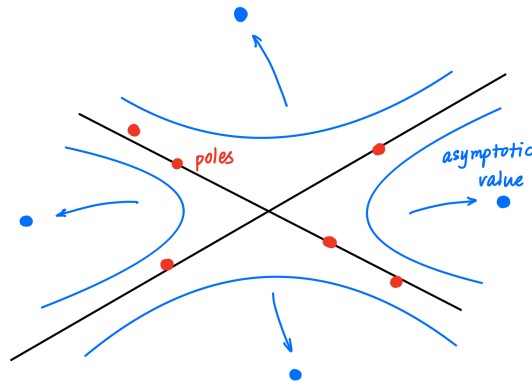
2.3.3 Remarks

Def (Nevanlinna function) $f \in S$, f be meromorphic function with polynomial Schwartz derivative.

Schwartz derivative is defined as

$$S(f) = \left(\frac{f''}{f'}\right)' - \frac{1}{2}\left(\frac{f''}{f'}\right)^2.$$

Nevanlinna function can be considered as a generalization of $\tan z$. The poles are distributed near finitely many rays, and asymptotic values are in regions between the rays.



Question Will the range of dimension improve when the condition shrinks from class B to class S ?

2.4 Hausdorff dimension of escaping sets in class S

2.4.1 Sketch of the proof

Thm Let f be a meromorphic function, then

$$\{\dim_H I(f) : f \in S\} = [0, 2].$$

i.e. $\forall d \in [0, 2], \exists f \in S$, s.t. $\dim_H I(f) = d$.

Rmk We don't have such conclusion for entire functions. Mero. are easier because we can take advantage of poles.

Proof (sketch) We will use q.c. surgery to glue 2 Weierstrass $\wp$ functions. Then by making the period τ larger, the number of poles becomes smaller, means the growth order becomes smaller by Teichmüller's lemma, so the $\dim_H I(f)$ will grow.

2.4.2 q.c. surgery

Consider 2 Weierstrass $\wp$ functions $\wp_1, \wp_2$, with lattice

$$\Lambda_1 = \{m + n\tau_1\}, \Lambda_2 = \{m + n\tau_2\}$$

and critical values

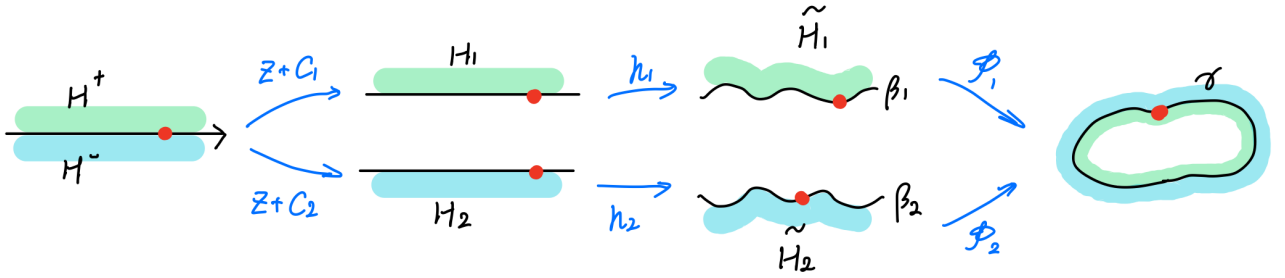
$$v_1 = \wp_1(0) = \infty, v_2 = \wp_1\left(\frac{1}{2}\right), v_3 = \wp_1\left(\frac{\tau_1 + 1}{2}\right), v_4 = \wp_1\left(\frac{\tau_1}{2}\right),$$

correspondingly, $w_1 = \wp_2(0) = \infty, w_2$ to w_4 so on.

If we put $\wp_1$ on the upper half plane H^+ and $\wp_2$ on the lower half plane H^- , we will find discontinuity on real axis, so we cannot use real axis as the edge to glue directly. However, we may modify the edge by some q.c. mapping, so that the image under $\wp_1, \wp_2$ fits well.

The whole process is illustrated as below. $\wp_1(h_1(z + c_1))$ maps H^+ to the inner part of a Jordan curve γ , while $\wp_2(h_2(z + c_2))$ maps H^- to the outside part. For $x \in \mathbb{R}$,

$$\wp_1(h_1(x + c_1)) = \wp_2(h_2(x + c_2)).$$



More specifically, given $\wp_1, \wp_2$, we can choose γ on the image plane first, then pull back to get the edges β_1, β_2 , which will decide c_1, c_2 . Denote

$$H_1 = \{z : \text{Im}(z) > \text{Im}(c_1)\}, H_2 = \{z : \text{Im}(z) < \text{Im}(c_2)\}.$$

Also denote $\tilde{H}_1$ as the region above β_1 , and $\tilde{H}_2$ the region below β_2 . We will finally show that there exist q.c. mappings

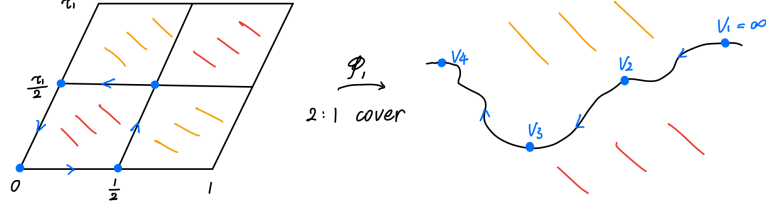
$$h_1(H_1) = \tilde{H}_1, h_2(H_2) = \tilde{H}_2.$$

Remember the whole process is to construct a mapping with controllable poles. Indeed, since we only modified a strip near real axis, all poles outside the strip are just part of Λ_1 and Λ_2 .

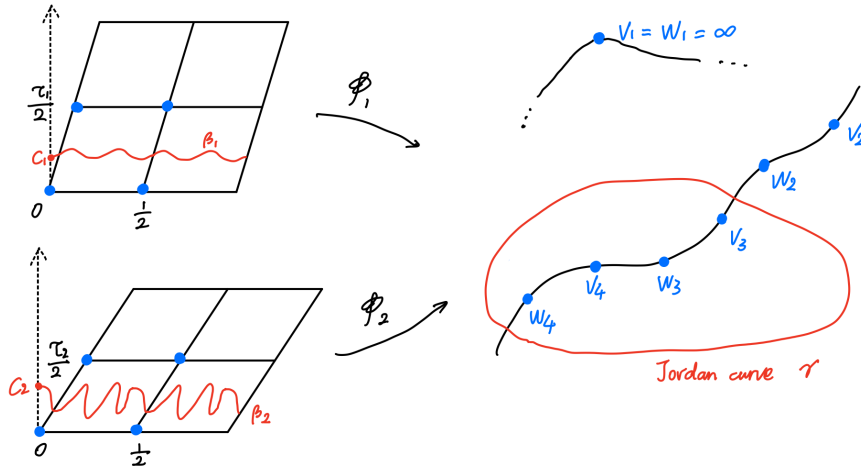
We begin our details with how Weierstrass $\wp$ function maps the grid to $\hat{\mathbb{C}}$. The close route

$$0 \rightarrow \frac{1}{2} \rightarrow \frac{1 + \tau_1}{2} \rightarrow \frac{\tau_1}{2} \rightarrow 0$$

is mapped to a curve which cuts $\hat{\mathbb{C}}$ to two regions. (The curve is simple, because there is no smaller period inside the grid.) By investing the direction, we mark the region by two colors to show this is a 2-to-1 cover.



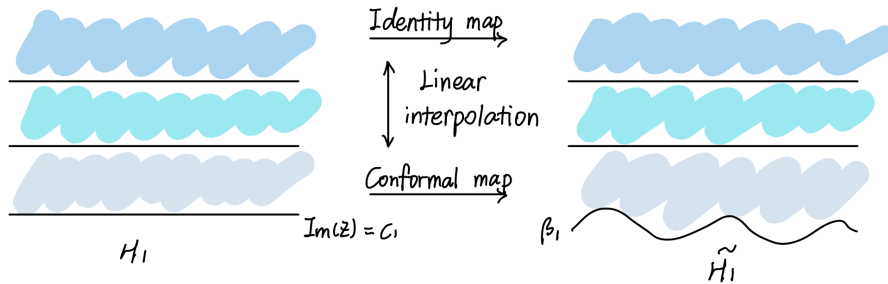
If we consider $\wp_1$ and $\wp_2$ at the same time, then they have 7 critical values in total. It is possible to choose a Jordan curve γ , such that v_3, w_3, v_4, w_4 are inside, and do not intersect the image of horizontal edges of the $1/4$ grid (refer to the figure below). Then pull back γ under $\wp_1$ to get β_1 , which is a curve lies in $\{0 < \text{Im}(z) < \text{Im}(\tau_1/2)\}$. Denote the region above β_1 by $\tilde{H}_1$, which looks like a half plane with curving edge.



β_1 may intersect the image axis many times, but we randomly choose one, denoted by c_1 . We want to construct a map from H_1 to $\tilde{H}_1$, which

1. Keeps the continuity on the edge;
2. Keeps the poles in general;
3. Has nice analytical property.

So for the strip near the edge, we take conformal map by Riemann mapping theorem. For neighborhood of ∞ , we take identity map. And for the strip between, we use linear interpolation (线性插值). Since linear interpolation is q.c., the whole map is q.c.



Note that such q.c. is not unique. If we do this to both H_1 and H_2 (actually only one side

is enough), then we do have

$$\wp_1(h_1(x + c_1)) = \wp_2(h_2(x + c_2)).$$

i.e. the two $\wp$ function can be glued together.

Lastly, the q.c. part of the map is supported on a strip. By Teichmüller-Wittich-Belinskii theorem, the integral is on a strip, thus finite. Therefore the map has same order as $f(z) = z$ when $z \rightarrow \infty$.

2.4.3 Open questions

Let S_q denote functions with q singular value. $\exp(z)$ and $\tan(z)$ are the two types of S_2 , standing for ∞ is/isn't asymptotic. The best result is in S_3 :

- Bergweiler-崔巍巍 2022, 2024:

$$\{\dim_H J(f) : f \in S_3\} = (0, 2].$$

Question: $\forall q \geq 4$, do we have

$$\{\dim_H J(f) : f \in S_q\} = (0, 2]?$$

Question: $\forall q \geq 3$, f ETF, the followings are true for class B , but for class S , do we have

$$\{\dim_H I(f) : f \in S\} = [1, 2]?$$

$$\{\dim_H J(f) : f \in S\} = (1, 2]?$$

3 Other Notes (with AI)

3.1 Pesin Theory

Pesin theory is a fundamental part of the modern theory of dynamical systems, particularly for the study of **non-uniformly hyperbolic systems**. It provides tools to understand the statistical behavior of orbits and the geometric structure of invariant measures in systems with **positive Lyapunov exponents**.

Pesin theory provides the foundation for studying:

- **SRB (Sinai-Ruelle-Bowen) measures:** invariant measures describing the physical behavior of chaotic attractors.
- **Non-uniformly hyperbolic systems:** such as logistic maps at positive entropy parameters or certain billiard systems.
- **Measure-theoretic properties:** ergodicity, mixing, entropy, and statistical stability.

3.1.1 Basic Setup

Let $f : M \rightarrow M$ be a $C^{1+\alpha}$ diffeomorphism on a compact Riemannian manifold M , and let μ be an f -invariant probability measure.

The **metric (or measure-theoretic) entropy** $h_\mu(f)$ quantifies the average rate of information production by the system with respect to the measure μ .

- By Oseledets' Multiplicative Ergodic Theorem, for μ -almost every point $x \in M$, there exists a splitting of the tangent space:

$$T_x M = E_x^1 \oplus \cdots \oplus E_x^k$$

such that for each nonzero $v \in E_x^i$,

$$\lambda_i(x) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \|Df_x^n v\|$$

exists and defines a **Lyapunov exponent**.

- A point x satisfying this decomposition and the limit is called a **regular point**.

3.1.2 Pesin Manifolds

For regular points with some positive and negative Lyapunov exponents, Pesin theory constructs local invariant manifolds:

- The **stable manifold** $W_{\text{loc}}^s(x)$: tangent to the direct sum of subspaces with negative exponents.
- The **unstable manifold** $W_{\text{loc}}^u(x)$: tangent to the direct sum of subspaces with positive exponents.

These manifolds vary measurably with x and are smooth embedded disks.

3.1.3 Pesin Entropy Formula

A key result of Pesin theory is the relation between metric entropy and Lyapunov exponents, known as the **Pesin Entropy Formula**:

$$h_\mu(f) = \int_M \sum_{\lambda_i(x) > 0} \lambda_i(x) \cdot \dim(E_x^i) d\mu(x).$$

This tells us that the entropy of the system with respect to μ equals the average rate of volume expansion in the directions where the system is expanding (i.e., along the positive Lyapunov exponents).

- The formula links two seemingly different notions: **local differential geometry** (via Lyapunov exponents) and **global statistical complexity** (via entropy).

- It provides a tool to compute entropy in smooth systems, especially when explicit partitioning is difficult.
- It implies that positive entropy systems must have positive Lyapunov exponents on a set of positive measure.
- It plays a central role in the study of **SRB measures**, which describe physical invariant measures for many chaotic systems.

Remark: If all Lyapunov exponents are non-positive almost everywhere, the entropy is zero.

3.1.4 Ruelle Inequality

Let $f : M \rightarrow M$ be C^1 and μ an invariant measure. Then:

$$h_\mu(f) \leq \int_M \sum_{\lambda_i(x) > 0} \lambda_i(x) \cdot \dim(E_x^i) d\mu(x).$$

This inequality is always valid under mild assumptions. The Pesin Entropy Formula can be seen as a refinement that characterizes when equality is attained.

3.2 Multiplicative Ergodic Theorem (Oseledets' Theorem)

The Multiplicative Ergodic Theorem (MET), due to V.I. Oseledets (1968), describes the asymptotic behavior of the derivative cocycle of a dynamical system. It is a generalization of the classical spectral theorem for matrices to the setting of dynamical systems.

In short, MET gives a precise description of how tangent vectors grow or contract under iteration of a differentiable map, and it forms the basis for defining **Lyapunov exponents**.

3.2.1 Basic Setup

Let:

- $f : M \rightarrow M$ be a measurable transformation of a probability space $(M, \mathcal{B}, \mu)$.
- f is μ -**invariant** and **ergodic**.
- $A : M \rightarrow \text{GL}(d, \mathbb{R})$ be a measurable matrix-valued map such that:

$$\int_M \log^+ \|A(x)\| d\mu(x) < \infty,$$

where $\log^+ t = \max\{\log t, 0\}$.

We study the **cocycle** generated by:

$$A^n(x) = A(f^{n-1}(x)) \cdots A(f(x))A(x).$$

3.2.2 Theorem (Oseledets, 1968)

Under the above assumptions, there exist:

- real numbers (Lyapunov exponents) $\lambda_1 > \cdots > \lambda_k$,
- and a measurable splitting of $\mathbb{R}^d$ into Oseledets subspaces:

$$\mathbb{R}^d = E_x^1 \oplus \cdots \oplus E_x^k,$$

defined for μ -almost every $x \in M$, such that:

1. For all $v \in E_x^i \setminus \{0\}$,

$$\lim_{n \rightarrow \pm\infty} \frac{1}{n} \log \|A^n(x)v\| = \lambda_i.$$

2. The subspaces E_x^i depend measurably on x , and are invariant under the cocycle:

$$A(x)E_x^i = E_{f(x)}^i.$$

3. If μ is ergodic, then the Lyapunov exponents λ_i and their multiplicities (dimensions of E_x^i) are constant μ -almost everywhere.

Geometrically, each Lyapunov exponent λ_i measures the exponential rate of growth (or contraction) of vectors in the corresponding subspace E_x^i . This gives a geometric decomposition of the tangent space into directions with different expansion/contraction behavior.

Also, Lyapunov exponents are symmetric under inversion:

$$\lambda \text{ is a Lyapunov exponent for } Df \iff -\lambda \text{ is a Lyapunov exponent for } Df^{-1}.$$

And the number of exponents (counting multiplicity) equals the dimension of the manifold.

3.3 Lyapunov-Type Quantities for Infinite-Dimensional Systems

In finite-dimensional systems, the Multiplicative Ergodic Theorem (MET) yields a finite list of Lyapunov exponents. In infinite-dimensional or noncompact settings, classical MET may not apply, but related notions exist. This subsection introduces three important generalizations or analogues:

1. Principal (or top) Lyapunov exponent,
2. Exponent of noncompactness,
3. Kuratowski measure of noncompactness.

3.3.1 Principal Lyapunov Exponent

Let $A^n(x)$ be a linear cocycle acting on a Banach space X . The **principal Lyapunov exponent** (also called the **top Lyapunov exponent**) is defined by:

$$\lambda_{\max}(x) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log \|A^n(x)\|.$$

This measures the **maximum exponential rate of growth** of the norm under the action of the cocycle. If A^n comes from the derivative cocycle of a dynamical system f , this corresponds to the maximal rate at which volumes expand (locally).

Remark: This quantity always exists (via subadditive ergodic theorem) and coincides with the largest Lyapunov exponent in finite dimensions.

3.3.2 Exponent of Noncompactness

In infinite-dimensional settings, the growth of the operator norm may be dominated by compact directions. Hence, we consider a weaker quantity called the **exponent of noncompactness**, which tracks the expansion in “essential” directions.

Let $\alpha(T)$ denote the **Kuratowski measure of noncompactness** of an operator T . Then the exponent of noncompactness is:

$$\chi(x) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log \alpha(A^n(x)).$$

This measures the exponential growth rate of the noncompact part of the cocycle A^n . It can be strictly less than the principal Lyapunov exponent and is often zero when the system is compactifying.

3.3.3 Kuratowski Measure of Noncompactness

Let $B \subset X$ be a bounded set in a Banach space. The **Kuratowski measure of noncompactness** $\alpha(B)$ is defined as:

$$\alpha(B) := \inf\{r > 0 : B \text{ can be covered by finitely many balls of radius } r\}.$$

For a bounded linear operator $T : X \rightarrow X$, define:

$$\alpha(T) := \alpha(T(B)),$$

where B is the closed unit ball in X . This gives a way to measure how far the operator is from being compact:

- $\alpha(T) = 0$ if and only if T is a compact operator.
- $\alpha(T) > 0$ indicates that T has genuinely noncompact behavior.

3.4 Arnold 直化定理 (Arnold Linearization Theorem)

在研究动力系统特别是保守系统的局部行为时，**Arnold 直化定理**提供了非线性系统在某些条件下可以被**线性系统**局部等价的情形。这个定理可以看作是对经典 Poincaré 定理的推广，并是 KAM 理论的重要前奏。

定理 (Arnold, 1963):

设哈密顿系统在平衡点附近可表示为：

$$\dot{\theta} = \omega + f(\theta, I), \quad \dot{I} = g(\theta, I),$$

其中 $(\theta, I) \in \mathbb{T}^n \times \mathbb{R}^n$ ， f, g 是足够光滑（如 C^∞ ）的函数，且在 $I = 0$ 处有：

$$f(\theta, 0) = 0, \quad g(\theta, 0) = 0.$$

若频率向量 $\omega \in \mathbb{R}^n$ 满足**非共振条件**，即对任意 $k \in \mathbb{Z}^n \setminus \{0\}$ 有：

$$\langle k, \omega \rangle \neq 0,$$

则存在一个 C^∞ 的微分同胚变换：

$$(\theta, I) \mapsto (\theta', I'),$$

使得系统在原点附近被**光滑地线性化**为：

$$\dot{\theta}' = \omega, \quad \dot{I}' = 0.$$

此定理说明在非共振频率条件下，哈密顿系统在某些共形变换下可**局部直化**。实质上，系统的轨道结构与其线性部分保持**拓扑等价**，并且保持了保守性质（如辛结构）。该定理为进一步研究小扰动下的系统稳定性（如 KAM 理论）奠定基础。

如果频率向量 ω 是 **Diophantine** 类型（满足一定的数论条件），则可获得更强的正则性（甚至解析）。对于共振情形，一般不能线性化，轨道结构会更加复杂，可能出现共振扭结、破坏稳定环面等现象。

3.5 Fine Focus and Weak Focus

Consider a planar system described by

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x} = (x, y) \in \mathbb{R}^2,$$

with an equilibrium at the origin $\mathbf{x} = \mathbf{0}$. Linearizing around this equilibrium gives

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x} + \mathbf{g}(\mathbf{x}),$$

where A is the Jacobian matrix at the origin, and $\mathbf{g}(\mathbf{x})$ contains higher-order terms.

3.5.1 Fine Focus

A **fine focus** is a type of equilibrium point where the linear part A has a pair of complex conjugate eigenvalues with nonzero real part:

$$\lambda_{1,2} = \alpha \pm i\beta, \quad \alpha \neq 0, \beta \neq 0.$$

The trajectories near the equilibrium spiral either towards it if $\alpha < 0$ (stable focus), or spiral away if $\alpha > 0$ (unstable focus). The nonzero real part ensures a **clear exponential rate** of attraction or repulsion.

3.5.2 Weak Focus

A **weak focus** arises when the linearized system has purely imaginary eigenvalues:

$$\lambda_{1,2} = \pm i\beta, \quad \beta \neq 0,$$

and the stability of the equilibrium cannot be determined from the linear part alone because the real part is zero ($\alpha = 0$). Here, the nonlinear terms $\mathbf{g}(\mathbf{x})$ govern the local dynamics.

In this case, the system may exhibit a **center** (closed orbits around the equilibrium) or a weak focus where the trajectories spiral very slowly either towards or away from the equilibrium, depending on higher-order terms. This behavior is more subtle and requires techniques such as normal form theory or Lyapunov quantities for analysis.

3.6 Huffman Information Theorem

The **Huffman Information Theorem** is a fundamental result in information theory that provides a method to construct an optimal prefix code for a set of symbols with known probabilities. This code minimizes the expected codeword length, achieving the best possible compression for lossless encoding.

3.6.1 Basic Setup

Consider a discrete memoryless source emitting symbols from an alphabet $\mathcal{A} = \{a_1, a_2, \dots, a_n\}$ with probabilities

$$P = \{p_1, p_2, \dots, p_n\}, \quad \sum_{i=1}^n p_i = 1.$$

We are going to construct a binary prefix code (codewords with no codeword being a prefix of another) that minimizes the average codeword length:

$$L = \sum_{i=1}^n p_i l_i,$$

where l_i is the length of the codeword assigned to symbol a_i .

3.6.2 Huffman's Algorithm and Theorem

Huffman's algorithm produces such an optimal prefix code. The theorem states:

For any discrete memoryless source with symbol probabilities P , the Huffman code yields a prefix code with minimum average codeword length L .

Moreover, the average code length L satisfies

$$H(P) \leq L < H(P) + 1,$$

where $H(P)$ is the Shannon entropy of the source:

$$H(P) = - \sum_{i=1}^n p_i \log_2 p_i.$$

3.7 Basin of an Invariant Measure μ

Let $(X, \mathcal{B}, f)$ be a dynamical system, where X is a metric space, $\mathcal{B}$ is the Borel σ -algebra, and $f : X \rightarrow X$ is a measurable map. Let μ be an f -invariant Borel probability measure, i.e.,

$$\mu(f^{-1}(A)) = \mu(A) \quad \text{for all } A \in \mathcal{B}.$$

The **basin of the measure** μ , often denoted by $B(\mu)$, is defined as the set of all points $x \in X$ for which the time averages of the empirical measures converge weakly to μ :

$$B(\mu) := \left\{ x \in X : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \delta_{f^i(x)} = \mu \right\}.$$

Here, δ_y denotes the Dirac measure concentrated at y , and convergence is in the weak-* topology on the space of probability measures on X .

The basin $B(\mu)$ consists of the points whose orbits are **statistically representative** of the measure μ . In other words, the long-term time averages for these points reflect the spatial distribution encoded by μ .

If μ is **ergodic**, then by the **Birkhoff Ergodic Theorem**, $\mu(B(\mu)) = 1$, i.e., almost every point (with respect to μ) lies in its own basin.

Basins may have complicated structures, and for non-ergodic μ , the basin may have measure zero or not be measurable at all.

Basins are essential for understanding the observable dynamics of a system: only points in the basin of a measure μ can empirically see μ through time averages.

3.8 Weak Expanding Property

3.8.1 Definition

We say that a map f satisfies the **weak expanding property** if for a positive Lebesgue measure set of points $x \in M$, the Lyapunov exponent in at least one direction is positive, or

more concretely,

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \log \|Df(f^i(x))^{-1}\|^{-1} > 0.$$

Equivalently, this means that the average rate of expansion along orbits of x is positive:

$$\frac{1}{n} \sum_{i=0}^{n-1} \log \|Df(f^i(x))\| \rightarrow \text{positive}.$$

3.8.2 Interpretation

The weak expanding property allows for non-uniform behavior —the system may not expand at every iterate or in all directions, but on average there is **asymptotic expansion** along orbits.

This property is weaker than **uniform expansion**, which requires $\|Df(x)v\| \geq \lambda\|v\|$ for all x and some $\lambda > 1$.

It is often used in the study of SRB (Sinai–Ruelle–Bowen) measures and physical measures in non-uniformly hyperbolic systems.

3.9 Stability

3.9.1 Lyapunov Stability (Orbit Stability)

Let $f : M \rightarrow M$ be a dynamical system on a metric space M . A point $x \in M$ is said to be **Lyapunov stable** if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$d(x, y) < \delta \quad \Rightarrow \quad d(f^n(x), f^n(y)) < \varepsilon \quad \text{for all } n \geq 0.$$

This is a local notion: small perturbations of initial conditions lead to orbits that remain close for all future time.

3.9.2 Structural Stability

A dynamical system $f : M \rightarrow M$ is said to be **structurally stable** if its qualitative behavior persists under small C^1 perturbations. Formally, f is structurally stable if for every C^1 -close map g , there exists a homeomorphism $h : M \rightarrow M$ such that

$$h \circ f = g \circ h,$$

i.e., f and g are **topologically conjugate**. Structural stability implies a very strong form of robustness.

3.9.3 Statistical Stability

Let $f_\varepsilon : M \rightarrow M$ be a family of dynamical systems depending smoothly on a parameter ε , and suppose that each f_ε admits an f_ε -invariant measure μ_ε (e.g., an SRB or physical measure).

We say that f_0 is **statistically stable** if

$$\mu_\varepsilon \xrightarrow[\varepsilon \rightarrow 0]{\text{weak-}^*} \mu_0.$$

In other words, the invariant measures vary continuously under perturbations. This ensures that the **statistical properties** of the system (such as time averages and correlations) are robust.

3.9.4 Stochastic Stability

In stochastic stability, the system is perturbed by random noise. Consider a family of randomly perturbed systems (e.g., a Markov process or random map), with associated stationary (invariant) measures μ_ε .

A deterministic system f is said to be **stochastically stable** if, under small random perturbations, the stationary measures μ_ε converge to an invariant measure μ_0 of f :

$$\mu_\varepsilon \xrightarrow[\varepsilon \rightarrow 0]{\text{weak-}^*} \mu_0.$$

This is important in physical and applied contexts where systems are subject to small random noise.

3.9.5 Other Notions

- **Topological Stability:** f is topologically stable if all sufficiently close maps (in C^0 topology) are semi-conjugate to f .
- **Expansive Stability:** Some systems are expansive, meaning that distinct orbits diverge at an exponential rate —stability can be discussed in the presence of this property.
- **Numerical Stability:** If every approximate trajectory (pseudo-orbit) can be shadowed by a true orbit, the system exhibits a form of numerical stability.

3.10 Shishikura's Inequality

Let:

N_{attr} = Number of attracting cycles,

N_{para} = Number of parabolic cycles,

N_{Siegel} = Number of Siegel disks,

N_{Herman} = Number of Herman rings.

Shishikura inequality (1987):

$$N_{\text{attr}} + N_{\text{para}} + N_{\text{Siegel}} + 2N_{\text{Herman}} \leq 2d - 2.$$

3.10.1 Remarks

- The term $2d - 2$ corresponds to the number of critical points of a degree- d rational map (counted with multiplicity).
- Each attracting, parabolic, or Siegel cycle must contain at least one critical point in its basin of attraction or invariant domain.
- Each Herman ring must contain at least two critical points in its grand orbit. This explains the coefficient 2 before N_{Herman} in the inequality.
- This bound plays a central role in the classification of Fatou components and the study of rational dynamics.

3.11 Peano and Atomic Decompositions

3.11.1 Peano Decomposition

Let $X \subset \hat{\mathbb{C}}$ be a compact, connected set (i.e., a continuum). A **Peano decomposition** of X is a decomposition of X into disjoint subcontinua such that the resulting quotient space is a Peano continuum.

A **Peano continuum** is a compact, connected, locally connected metric space. i.e., it is the continuous image of the closed unit interval $[0, 1]$. Equivalently, it is a locally connected continuum.

In complex dynamics, the Peano decomposition is used to simplify the structure of a Julia set $J(f)$, especially when it is not locally connected. This allows one to model the dynamics on a simpler space that preserves enough topological and dynamical information.

3.11.2 Atomic Decomposition

The **atomic decomposition** is a finer notion, often used when studying the decomposition of a Julia set into its **indecomposable subcontinua**.

A continuum X is called **indecomposable** if it cannot be expressed as the union of two proper subcontinua. An atomic decomposition decomposes X into the minimal (with respect to inclusion) subcontinua which still capture its nontrivial topological structure.

The decomposition is upper semi-continuous, meaning the decomposition varies continuously in a certain topological sense.

3.12 Baby Mandelbrot Sets

A **Baby Mandelbrot set** refers to a smaller, self-similar copy of the Mandelbrot set that appears inside $\mathcal{M}$. These sets arise from the **tuning** process and exhibit **renormalization** behavior.

More precisely, baby Mandelbrot sets appear as copies of $\mathcal{M}$ near **Misiurewicz points** and **hyperbolic components** in $\mathcal{M}$, where the local dynamics can be described by a polynomial-like map that is conjugate to a quadratic polynomial.

3.12.1 Key Properties

- Each baby Mandelbrot set corresponds to a **renormalizable** parameter c , i.e., the dynamics of f_c on a suitable subset is topologically conjugate to a lower-degree map (usually again quadratic).
- The **Douady-Hubbard Straightening Theorem** ensures that every such polynomial-like map is hybrid equivalent to a true polynomial, this is how baby Mandelbrots are formally defined.
- The baby Mandelbrot sets are often attached to the main cardioid or other hyperbolic components, and they often resemble $\mathcal{M}$ itself in both shape and bifurcation structure.

3.12.2 Tuning

Tuning is a method introduced by Douady and Hubbard to describe how baby Mandelbrot sets are embedded in the main set.

- Tuning takes a copy of $\mathcal{M}$ and glues it into the parameter space using a combinatorial rule (based on external angles).
- Every baby Mandelbrot has a well-defined **tuning operator**, and the dynamics within the baby copy reflects the combinatorics of the larger set, but with periodic structure.

4 Appendix

4.1 List of lecture notes

- 任浩杰-The fractal dimension of graphs of Weierstrass type
- 崔贵珍-Rational maps with constant Thurston pullback mapping
- 文志英-分形的历史、现状和发展
- 曾劲松-有理函数动力系统简介

- 沈维孝-一致扩张映射上遍历优化问题的研究
- 胡俊-hand write lecture note
- 赵云-热力学机制和维数理论
- 邱维元-Boundaries of Hyperbolic Components of Cubic Polynomials

4.2 Curriculum

<http://maths.nju.edu.cn/~chengwei//SS/>

教学楼102					
	8:00-9:50	10:10-12:00	14:00-15:00	15:00-16:00	16:00-17:00
6.30 Mon.	陈国璋 天体力学	胡俊 平方&立方动力系统		程崇庆 我学动力系统的一点体会	黄文 Multiple Recurrence without commutativity
7.1 Tue.	胡俊 平方&立方动力系统			杨大伟 Ergodic theory of surface diffeomorphisms	
7.2 Wed.	周麒 准周期薛定谔算子	陈国璋 天体力学		段华贵 Maslov-type index theory and periodic orbits	王婧 刘维尔频率的拟周期系统的KAM理论及其应用
7.3 Thu.	连增 Lyapunov指数, 乘积遍历定理	陈和柏 高阶退化奇点	连增 Lyapunov指数, 乘积遍历定理		杜增吉 Dynamics of the reaction-diffusion models in advective
7.4 Fri.	陈和柏 高阶退化奇点	周麒 准周期薛定谔算子		曹永罗 Physical measures for a class of partially hyperbolic attractors	王林 扭转映射不变圈的动力学
7.7 Mon.	曾劲松 有理函数动力系统	崔巍巍 逃逸集维数	沈维孝 一致扩张映射上遍历优化问题的研究	邱维元 Boundaries of hyperbolic components of cubic	甘少波 Lyapunov stable chain classes of singular flows
7.8 Tue.	崔巍巍 逃逸集维数	曾劲松 有理函数动力系统		崔贵珍 Rational maps with constant Thurston pullback mapping	罗俊 分形方块的拓扑结构与Lambda函数
7.9 Wed.	严军 Hamilton动力系统	张国华 可调群		尤建功 局域化、相变和迁移率边	张祥 Analytic normal form of integrable differential systems
7.10 Thu.	张国华 可调群	赵云 拓扑压, 热力学机制, 维数		胡锡俊 N体问题周期轨道简介	冀书关 Breather solutions for the nonlinear wave equation in
7.11 Fri.	赵云 拓扑压, 热力学机制, 维数	严军 Hamilton动力系统			

教学楼101			
	9:00-11:00	14:00-17:00	
6.30 Mon.	曹仕平-Construction of Dirichlet forms on Fractals 任浩杰-The Fractal Dimension of Graphs of Weierstrass-Type Functions		
7.1 Tue.		学员报告	
7.2 Wed.		任浩杰	
7.3 Thu.		特邀报告	
7.4 Fri.			
7.7 Mon.	丰德军-Dimension Estimates for C1 Iterated Function System		
7.8 Tue.			
7.9 Wed.		学员报告	
7.10 Thu.		特邀报告	
7.11 Fri.			